

Staff Task Dashboard & SLA Engine BRD

1. Summary

A mobile-first web application (and potential WhatsApp integration) designed for Vibe House staff to access on their personal smartphones. It digitizes, routes, and monitors all operational service requests, eliminating verbal coordination and ensuring service consistency through strict SLA enforcement and load-balanced assignment.

2. Task Lifecycle & Statuses

Every service request generated by the system (whether triggered by a guest's payment or a free request) moves through three distinct phases:

1. **Pending:** The task is created and auto-assigned to a specific staff member, but no one has acknowledged or picked it up yet.
2. **In-Progress:** The assigned staff member has clicked "Acknowledge/Pick," and the system updates the database. The staff is currently executing the task.
3. **Completed:** The staff member marks the task as finished, closing the loop and notifying the guest.

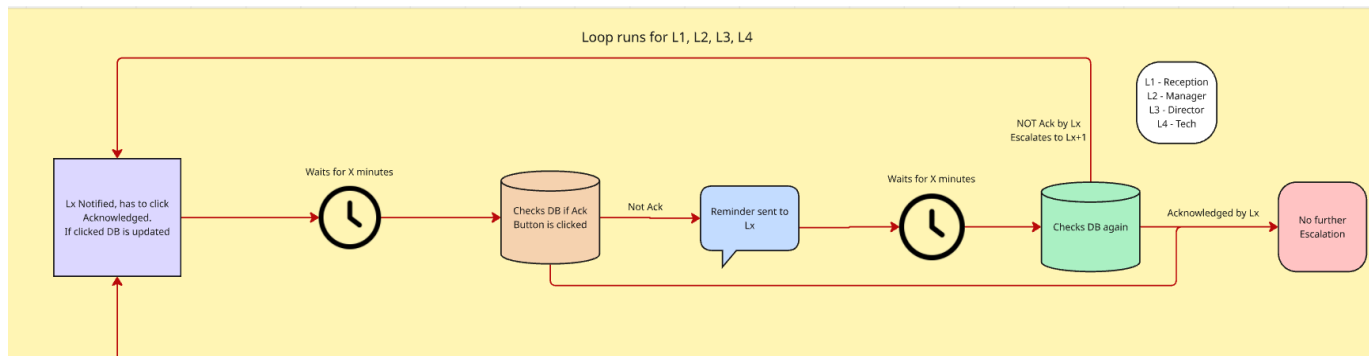
3. Auto-Assignment Logic (Load Balancing)

To prevent overwhelming any single staff member, the system uses a dynamic load-balancing algorithm.

- **Trigger:** A new service request is generated (e.g., "Deliver Bath Towel to Room 101").
- **Staff Query:** The system identifies all staff currently "Active/On-Shift" for that specific department (e.g., Housekeeping).
- **Load Calculation:** The system counts how many tasks currently have the status **In-Progress** for each active staff member.
- **Assignment:** The task is automatically assigned to the staff member with the minimum (**min**) number of **In-Progress** tasks.

4. SLA & Escalation Protocol

The system enforces a strict acknowledgment and escalation loop to ensure no task is ignored. Based on your escalation matrix, the loop runs sequentially through predefined levels (e.g., \$L_0\$: Assigned Staff, \$L_1\$: Reception, \$L_2\$: Manager, \$L_3\$: Director/Owner, \$L_4\$: Tech).\



The Escalation Loop (\$L_x\$ to \$L_{x+1}\$):

1. **Initial Notification:** Level \$L_x\$ is notified of the task and must click the "Acknowledge" button.
2. **Timer 1:** The system waits for \$X\$ minutes.
3. **DB Check 1:** The system checks the DB to see if the "Acknowledge" button was clicked.
 - *If Acknowledged:* The DB is updated to **In-Progress**, and no further escalation occurs.
 - *If Not Acknowledged:* A "Reminder" notification is sent to \$L_x\$.
4. **Timer 2:** The system waits for another \$X\$ minutes.
5. **DB Check 2:** The system checks the DB again.
 - *If Acknowledged:* No further escalation occurs.
 - *If Not Acknowledged:* The task automatically escalates to \$L_{x+1}\$ (the next level up), and the exact same notification loop restarts for the higher authority.

5. Exception Handling: SLA Pause Requests

Staff members may encounter legitimate delays (e.g., a room is locked, or a spare part is needed) that prevent them from completing an **In-Progress** task within the allotted time.

- **Request:** The staff member clicks "Request SLA Pause" on their interface and selects a reason from a dropdown.
- **Approval Flow:** A notification is instantly sent to their upper-level supervisor (e.g., Manager).
- **Action:** The Manager must review and click "Approve" or "Reject".
 - *Approved:* The SLA timer freezes, preventing an artificial breach/escalation.
 - *Rejected:* The SLA timer continues running, and the staff must complete the task immediately.

6. User Interfaces (Mobile Web-App / WhatsApp)

Since staff will use their personal smartphones, the UI will be lightweight, high-contrast, and optimized for touch.

A. Staff View (L0/L1):

- **Task Feed:** A simple queue showing Assigned Tasks.
- **Card Details:** Room number, Guest name, Task type (e.g., "Hot Water Issue"), and priority tag (TBD rules).
- **Action Buttons:** Prominent "Acknowledge" and "Mark Completed" buttons.
- **Exception Buttons:** "Request Pause".
- **Countdown:** A visible SLA timer showing remaining time before a breach/escalation.

B. Supervisor/Manager View (\$L_2\$ and above):

- **Global Queue:** Visibility of all pending and in-progress tasks across the property.
- **Escalation Alerts:** High-priority visual flags for tasks that have escalated to their level due to \$L_0/L_1\$ timeout.
- **Pause Approvals:** A dedicated section to approve/deny staff pause requests.
- **Metrics:** A quick view of department bottlenecks and staff productivity tracking (e.g., who is consistently missing SLAs).